1st SIT Coursework - 01 Question paper **Year Long 2025/2026**

Module Code:	**CS5002NT**
Module Title:	**Software Engineering**
Module Leader:	**Mr. Rubin Thapa (Islington College)/Mr. Amit Shrestha**

Coursework Type:	Groupwork
Coursework Weight:	This coursework accounts for ***20%*** of the overall module grades.
Submission Date:	Week 13 (**January 19th, 2026**) First Milestone: **December 22nd, 2025** Second Milestone: **January 5th, 2026**
Coursework given out:	Week 6
Submission Instructions:	Submit the following to the Islington College's MST portal **before 09:00 PM** on the due date: ● A report (document) in .pdf format in the MST portal or through any medium which the module leader specifies.
Warning:	London Metropolitan University and Islington College take plagiarism very seriously. Offenders will be dealt with sternly.

PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University *Regulations on Cheating, Plagiarism and Collusion*

Section 2.3:*"The following broad types of offence can be identified and are provided as indicative examples ….*

(i) Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.

(ii) Falsifying data in experimental results.

(iii) Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.

(iv) Bribery or attempted bribery of a person thought to have some influence on the candidate's assessment.

(v) Collusion to present joint work as the work solely of one individual.

(vi) Plagiarism, where the work or ideas of another are presented as the candidate's own.

(vii) Other conduct calculated to secure an advantage on assessment.

(viii) Assisting in any of the above.

Some notes on what this means for students:

1. Copying another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.

2. Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. "$e = mc^2$ (Einstein 1905)". A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

School of Computing, FLSC

CONTRACT CHEATING

Contract cheating (also known as assessment outsourcing, commissioning or ghost writing) is when someone seeks out another party, or AI generator service, to produce work or buy an essay or assignment, either already written or specifically written for them or the assignment to submit as their own piece of work.

Contract cheating undermines the integrity of the academic process and devalues the qualifications awarded by the university. Students are reminded that academic integrity is a fundamental principle of our institution. Engaging in contract cheating not only impacts the individual's academic record but also the reputation of the university.

Students are encouraged to seek support if they are struggling with their coursework. The university offers a range of resources, including academic counseling, tutoring services, and workshops on study skills and time management. Utilizing these resources can help students achieve their academic goals without resorting to dishonest practices.

Penalty:

- Failure in the Module: The student must re-register for the same module, and the re-registered module will be capped at a bare pass.

- Ineligibility to Continue on the Course: Where re-registration of the same module, or a suitable alternative, is not permissible, the student will not be able to continue on the course. Additionally, the following penalty will be applied to the student's final award:

    - Undergraduate Honors: The student's final classification will be reduced by one level.
    - Unclassified Bachelors: Downgraded to Diploma in Higher Education.
    - Foundation Degree: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Certificate in Higher Education.
    - Masters: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Postgraduate Diploma.

Reporting and Consequences:

Instances of contract cheating will be thoroughly investigated, and students found guilty will face the penalties outlined above. It is the responsibility of every student to ensure that their work is their own and to avoid situations that could lead to accusations of academic misconduct.

By adhering to these standards, students contribute to a fair and equitable academic environment, ensuring the value and recognition of their qualifications are maintained.

1. Introduction

This assignment contributes 20% to the overall mark for this module and involves group work. You are expected to form groups of 5 or 4 students.

2. Objectives

✓ To demonstrate practical knowledge of ‘Structured Software Engineering’(Yourdon)

✓ To submit some basic Project Management Artifacts.

✓ To work successfully in a small group to a given time scale.

3. Client Background

Gokyo Bistro is a popular restaurant located in Jhamsikhel, Lalitpur. It offers a refreshing ambience with both indoor and outdoor dining options. Recently, the restaurant has been facing challenges in its regular operations related to table arrangements and order management. There has been a significant surge in customers, and many have had to leave due to maximum occupancy. Additionally, there has been an increase in requests for home delivery. Therefore, the management is looking for an online system that can handle all these operations efficiently. Assuming you and your team have been approached for this project, you are required to carry out Structured Analysis and Design based on the specifications given below.

3.1 Detailed specification of GROUP task

1. ***Login and Registration***

   The system should have 3 types of users. Admin, visitors and members. Admin and Members can login with their respective credentials.

2. ***Book Table***

   Visitors/Members can check the availability of tables and reserve the table for a particular date/time. There is a requirement to pay 1200 advance in order to book the table.

3. ***Cancel Reservation***

   Reservation can be cancelled before the final day whereas in case the reservation is cancelled on the very day, a fee of 500 is deducted from advance before returning the amount to the visitors/members.

4. ***Set Beverage choices***

   Though not compulsory, while booking or even after booking, visitors can set the expected beverages choices they might prefer from the available list whereas members can reserve their wine/whiskey choices.

5. ***Place Order for Home Delivery***

   There should be an option to make order for home delivery for Members. Payment can be made online as well.

6. ***Generate Report***

   Admin should have a facility to generate the financial report. An advanced mechanism is needed for admin to predict the revenue for further years along with the recommendation of menu items.

7. ***Bill Generation***

   There should be the privilege to generate bills for admin and generate receipts.

8. ***Update Menu***

   Admins can add menu items, delete and update menu items.

9. ***Rate the experience***

   Visitors/members can rate the overall experience of food, staff's behavior , ambience and feedback can also be provided.

10. Post offers and schemes

Admin should be able to post various special offers and discount schemes from time to time.

4. Your Tasks

Your main tasks are:

4.1 Business Case

- Following proper template, create a business case that provides clear, concise justification for project initiative, ensuring that it aligns with organizational goals and delivers values.

4.2 SRS

- **Functional Requirements**
- **Non-Functional Requirements**
    - **Design and Implementation Constraints**
    - **External Interfaces Required**

      (User Interfaces, Hardware Interfaces, Software Interfaces, Communication Interfaces)
  - **Other Non-Functional requirements.**

4.3 Environmental model specification

- Context Level diagram

4.4 Internal model specification for the system.

- (DFD) Data Flow Diagram (level1(**for all features**), level2(**for any 3 processes which should not include features given in the individual task like Book table, Place Order for Home Delivery, Set Beverage Choices, Update Menu and Rate the experience**))
- Entity Relationship Diagram (ERD)

- Data Dictionary (with definitions of major data flows and definitions of data stores and entities)
- Process specifications (Pspecs) for elementary processes.

4.5 Design specification

- Structure chart (upper level) for the whole system.

4.6 Progress Logs.

You are asked to produce to track overall progress of the project:

- All the ***vagueness/inconsistencies*** you have discovered in the provided specification and the ***assumptions*** you made.
- A table comprising clearly written Students name, London met ID that shows the contribution in the group task, individual module chosen.
- Group meeting minutes that include the meeting agenda, items discussed, problems faced etc.

5. Detailed specification of INDIVIDUAL task

As the individual task you are asked to produce a number of analysis and design specifications of a particular part of the system.

Each group member should select ONE of the following 'functions' for their individual task and clearly specify **their name, London Met ID and their chosen functions** clearly.

1. Book Table

2. Rate the experience

3. Set Beverage Choices

4. Update Menu

5. Place Order for Home Delivery

After selecting a function create the design specifications of the system as mentioned below:

5.1 Environmental model specification

- Context Level

5.2 Internal model specification for the system

- The Level 1 DFD fragments
- Level 2 DFDs for the particular function

5.3 Design specification

- Structure chart for the particular function.
- Module specifications (MSpecs) for corresponding modules
  Please note that a template for MSpecs is given below under Footnotes.

Note:

Each group should produce only ONE integrated copy of the report with the following structure:

Cover page – Coursework title plus names and London Met id numbers of the student.

Table of Contents format

1. Introduction (a brief overview of the report)
2. Group task
   - 2.1Business Case
   - 2.2SRS
   - 2.3Context Level Diagram/Level 0 DFD
   - 2.4 DFD Level 1 - (**for all features**)
   - 2.5 DFD Level 2 - (**for any 3 processes which should not include features given in the individual task like Book Table, Place Order for Home Delivery, Set Beverage Choices, Update Menu and Rate the experience**)
   - 2.6 Entity Relationship Diagram (ERD)
   - 2.7 Data Dictionary (with definitions of major data flows and definitions of data stores and entities)
   - 2.8 Process specifications (Specs) for elementary processes.
   - 2.9 Design specification
     - Structure Chart
   - 2.10 Progress logs

3. Individual task

Student Name: Rubin Thapa
LMU ID: 2XXXXXX8
Function chosen: Place Order for Home Delivery

3.1 Context Level
3.2 Level 1 DFD fragments
3.3 Level 2 DFDs for the particular function
3.4 Design specification

- Structure chart for the particular function.
- Module specifications (MSpecs) for corresponding modules (Please note that a template for MSpecs is given below under Footnotes)

Student Name: Ishan Singh Thakuri
[“” Continue as above’’’]

Student Name: Sanjeep Lama
[“”” Continue as above’’’]

4 Conclusion (a brief summary of the report)
5 References

Foot-notes:

(MSpec) Component/Module specification (in individual task, should have the following format:)

MODULE NAME
PURPOSE. A brief description of the module's function
PSEUDOCODE. Detailed description of the module's function
INPUT PARAMETERS (if any)
OUTPUT PARAMETERS (if any)
GLOBAL VARIABLES (if any)
LOCAL VARIABLES (if any)
CALLS. References to other modules/components (including libraries).
CALLED BY. Names of other modules/components using this one.

Presentation’s viva (VIVA by each group member during each milestones week and final VIVA date will be notified later.)

Plagiarism and / or collusion. Any evidence of plagiarism or collusion between groups and / or with entities outside the university may result in an investigation by the case work office and, if substantiated, one of a range of serious penalties may be applied to the student[s] concerned.

Marking Scheme

Group Marks	Marks
Business Case	**7**
SRS	**10**
Context Diagram	**6**
DFD	**10**
E-R Diagram	**8**
Data Dictionary	**5**
Process Specification	**6**
Structured Chart	**6**
Progress Logs	**8**

Individual Marks	
Context Diagram	**5**
DFD	**6**
Structure Chart	**6**
Module Specification	**10**
Description	**7**

MILESTONES BREAKDOWN

MILESTONE 1 (Week 9)

- Introduction
- Business Case
- SRS
- DFD’s (level 0, level 1 and 2)
- E-R diagram
- Data Dictionary
- Process Specification

MILESTONE 2 (Week 11)

- Group Structure Chart
- Progress Logs

Individual Task:

- DFD’s (Level 0, Level 1 and 2)
- Structured Chart (Group and individual)
- Description of diagrams (for individual)
- Module Specification(individual)
- Conclusion

END